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Experimental study on dual recirculation of polymer electrolyte membrane fuel cell

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ABSTRACT

Durability and start-up ability in sub-zero environment are two technical bottlenecks of vehicular polymer electrolyte membrane (PEM) fuel cell systems. With exhaust gas recirculation on the anode and cathode side, the cell voltage at low current density can be reduced, and the membrane can be humidified without external humidifier. They may be helpful to prolong the working lifetime and to promote the start-up ability. This paper presents an experimental study on a PEM fuel cell system with anodic and cathodic recirculation. The system is built up based on a 10 kW fuel cell stack, which consists of 50 cells and has an active area of 261 cm^2. A cathodic recirculation pump and a hydrogen recirculation pump are utilized on the cathode and anode side, respectively. Key parameters, e.g., stack current, stack voltage, cell voltage, air flow, relative humidity on the cathode side, oxygen concentration at the inlet and outlet of the cathode side, are measured. Results show that: 1) with a cathodic recirculation the system gets good self-humidification effect, which is similar to that with an external humidifier; 2) with a cathodic recirculation and a reduction of fresh air flux, the cell voltage can be obviously reduced; 3) with an anodic recirculation the cell voltage can also be reduced due to a reduction in the hydrogen partial pressure, the relative humidity on the cathode side is a little smaller than the case with only cathode recirculation. It indicates that, for our stack the cathodic recirculation is effective to clamp cell voltage at low current density, and a self-humidification system is possible with cathodic recirculation. Further study will focus on the dynamic model and control of the dual recirculation fuel cell system.

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Introduction

Air pollution and energy crisis bring strict restrictions on vehicle emission and energy consumption. Polymer electrolyte membrane (PEM) Fuel Cell Electric Vehicle (FCEV) is one of the most promising vehicles in the future with the advantages of zero emission, high efficiency and the probability of using renewable energy, and it has been research hotspot for a long time [1,2]. However, there are still bottlenecks that block the rapid commercialization of FCEVs. Durability and start-up ability in sub-zero environment are two of them [2–4].

Durability is a big challenge in vehicular fuel cell applications. There are different factors that influent the working lifetime of a fuel cell. Among them, the high potential in idle or low load states is a key one. Previous researches showed that, operating in high potential state leads to a dissolution of Platinum catalyst and an corrosion of carbon material, thus it accelerates the degradation of fuel cell [5–8]. Experimental study by Noto et al. [9] showed that fuel cell performance degradation rate under potential cycling from 0.65 V to open circuit voltage (OCV) was more than 6 times than that from 0.65 V to 0.85 V. However, in most cases of automotive applications, idle and low-load states are ineluctable, because the output power of a fuel cell engine is determined by working conditions of the vehicle. Reducing cell voltage in low current density is therefore necessary in order to increase the lifetime of a fuel cell.

Inlet gas humidification has great impact on the performance of fuel cell, and it associates with the ionic conductivity of membrane [10]. So usually one or two external humidifiers is needed to avoid dehydration of membrane, which becomes an obstacle to fast star-up in sub-zero environment. When a fuel cell system is equipped with an external humidifier, it normally takes a long time to heat the water that is contained in the humidifier. Thus, it is an essential step to remove the external humidifier for shortening the start-up time.

Several studies have been conducted for technologies on self-humidification and voltage clamping. One possibility is to use exhaust gas recirculation systems. With exhaust gas recirculation on the anode or the cathode side, the cell voltage at low current density can be reduced, and the membrane can be humidified without an external humidifier. They may be helpful to prolong the working lifetime and to promote the start-up ability.

Exhaust gas recirculation (EGR) can be designed for the anode or the cathode system. A lot of researches on anodic recirculation of PEMFC have been carried out. Different types of recirculation system, including systems with recirculation pump [11,12], injection pump [13–15] or pumpless system [16,17], have been studied in order to improve fuel consumption and realize humidification of hydrogen. Toyota has developed a novel fuel cell system for its FCEV Mirai. Self-humidification is realized through H_2 recirculation and special cell structure [18].

In a fuel cell system with cathodic recirculation, part of the cathode exhaust gas, which has high temperature and high humidity, is re-circulated to be mixed with fresh air. The inlet gas in cathode is humidified and the oxygen concentration is reduced, which reduces the voltage of fuel cell at open circuit or low current conditions. Although simply reducing inlet air or hydrogen flow rate can also reduce cell voltage, nonuniformity of reactant distribution becomes severe. Breault et al. [19] and Thomas et al. [20] proposed different cathodic recirculation systems for inlet air humidification. Kim et al. [21] studied exhaust gas recirculation of fuel cell by theoretical and experimental methods, and validated its humiliation effect. Test with a 5-cell stack showed that, performance with an exhaust gas recirculation is a little less than that with a membrane humidifier. Hu et al. [22] studied the relationship between cathodic recirculation rate and the internal water content of fuel cell through modeling and simulation. Water management strategy is proposed based on cathodic recirculation. Cheng et al. [23] proposed an air supply strategy based on a cathodic EGR in order to keep cell voltage under 0.8 V and to improve the lifetime of fuel cell in the method of modeling and simulation.

From above it is known that, most of the recent studies related to gas recirculation of a fuel cell system focus on EGRs on one side, and are conducted theoretically. An experimental study for a system with dual recirculation is really rare. This paper presents an experimental study on a PEM fuel cell system with anodic and cathodic recirculation. A fuel cell system with a 10 kW, 50 cells, 261 cm^2 fuel cell stack is built up to realize anodic and cathodic recirculation, and key parameters are measured. Self-humidification and voltage clamping functions of cathodic recirculation are verified and the results are analyzed afterward.

The remaining content of this paper is organized as follow. Section System structure introduces the built fuel cell system. Active voltage clamping and self-humidification are theoretically analyzed in Section Theoretical analysis. Experimental study and results are presented in Section Experimental study. Conclusions are presented in Section Conclusion.

System structure

The experiments presented in this paper are conducted on a system based on a 10 kW PEMFC test platform. Physical appearance and schematic of the test system are shown in Figs.1 and 2, respectively. It consists of air supply system, hydrogen supply system, cooling system, cell voltage monitor and an electric load. The specifications of the testbench are listed in Table 1. The main function of the test bench is to control the flow rate of air and hydrogen. Humidification is also available on both anode and cathode sides. A membrane humidifier is utilized on the cathode side, and a bubble humidifier is utilized on the anode side. A cathodic recirculation pump is used to recycle part of the cathode exhaust gas to the cathode inlet, and a back-up pressure valve is used to control the cathode pressure. Hydrogen recirculation is built up in the same way as cathode. A purge valve and a bypass valve in the anode outlet provide different operation modes for the anode system (i.e. the direct through mode, the dead-end mode, and the recirculation mode). The cooling system keeps the stack temperature within a proper range, and the cell voltage monitoring system measures and records all the single cell voltages and the total voltage of the stack. In order to monitor

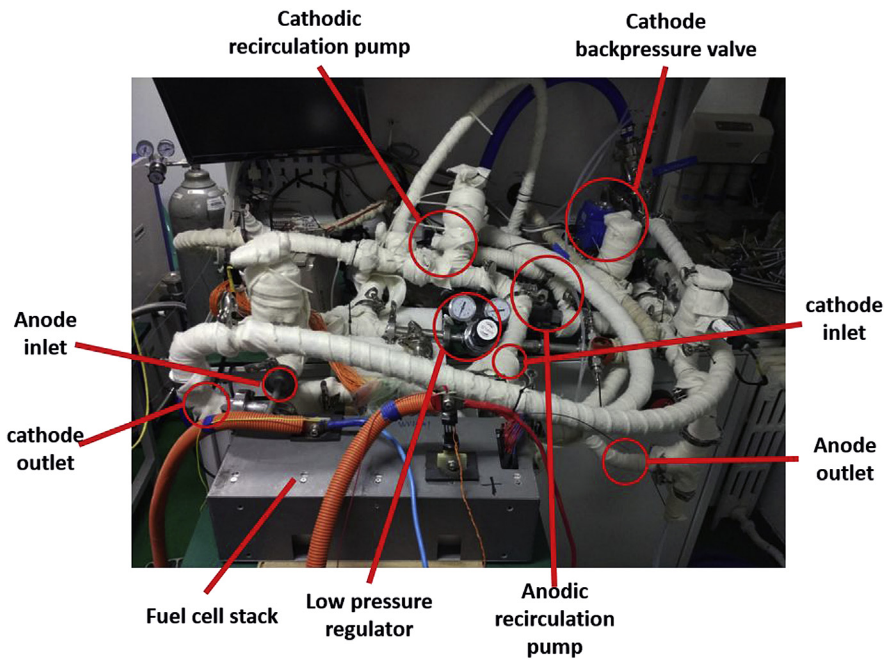

Fig. 1 – Physical appearance of the system.

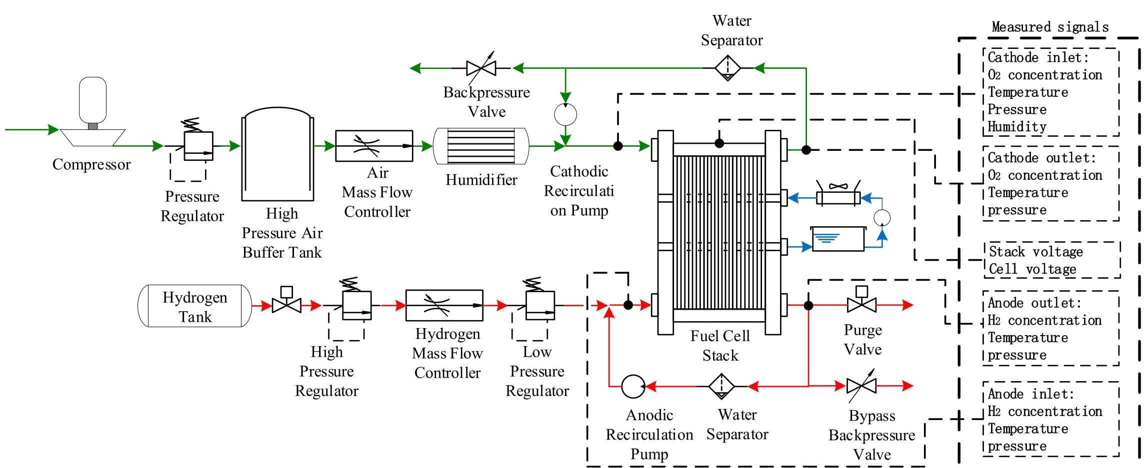

Fig. 2 – Schematic of the overall system.

Table 1 – Specifications of testbench.

Parameters	Values
Max test power	20 kW/600 V/600 A
Max flow rate (anode)	250 L/min
Max flow rate (cathode)	750 L/min
Max gas temperature (anode)	85 °C
Max gas temperature (cathode)	80 °C
Humidification (anode)	Bubble humidifier, dew point control
Humidification (cathode)	Membrane humidifier, dew point control

the relative humidity and oxygen concentration of inlet air, a humidity sensor is installed in the inlet of the cathode side, and two oxygen sensors are placed in the inlet and outlet of the cathode side, respectively.

Theoretical analysis

The theoretical analysis is based on the following assumptions:

1. The system works in a steady state.
2. Each gas is considered as an ideal gas.
3. Mole fraction of water in ambient air is relatively small so it is neglected. Other components of air except oxygen and nitrogen are also neglected.
4. Hydrogen fraction in gas from hydrogen tank is considered as 100%.
5. Nitrogen accumulation in anode is neglected since gas purging is applied in anode in this study, which removes condensed water and accumulated nitrogen.
6. Water condensation is neglected during gas mixture.

Part of the cathode exhaust gas, whose temperature and humidity are high, is recirculated and mixed with fresh air in the fuel cell system with cathodic recirculation. Eqs. (1)–(3) show the flow rate in mole/s of oxygen, nitrogen and water, $\dot{W}_{O_2,in,ca}$, $\dot{W}_{N_2,in,ca}$ and $\dot{W}_{vap,in,ca}$, in the inlet of cathode according to species conservation.

$$\begin{aligned}\dot{W}_{O_2,in,ca} &= W_{dry,ca}\cdot x_{O_2,dry,ca} + W_{re,ca}\cdot x_{O_2,out,ca}\\ &= W_{dry,ca}\cdot x_{O_2,dry,ca} + \alpha\cdot W_{ex,ca}\cdot x_{O_2,out,ca}\end{aligned} \tag{1}$$

$$\begin{aligned}\dot{W}_{N_2,in,ca} &= W_{dry,ca}\cdot x_{N_2,dry,ca} + W_{re,ca}\cdot x_{N_2,out}\\ &= W_{dry,ca}\cdot x_{N_2,dry,ca} + \alpha\cdot W_{ex,ca}\cdot x_{N_2,out,ca}\end{aligned} \tag{2}$$

$$\dot{W}_{vap,in,ca} = W_{re}\cdot x_{vap,out,ca} = \alpha\cdot W_{ex,ca}\cdot x_{vap,out,ca} \tag{3}$$

Here, $W_{dry,ca}$, $W_{ex,ca}$ and $W_{re,ca}$ stand for flow rate of dry air from compressor, exhaust air and recirculated exhaust air, respectively; $x_{i,dry,ca}$, $x_{i,\ out,\ ca}$ and $x_{i,in,ca}$ ($i = O_2$, N_2, vap) represent molar fractions of gas species in dry air, exhaust air and inlet air, respectively. Recirculation ratio, α, is defined as the ratio of the recirculated exhaust gas to the overall exhaust gas in cathode. The molar fraction of oxygen and water vapor in the cathode inlet gas can be derived from equations, as shown in following equation.

$$\begin{aligned}x_{O_2,in,ca} &= \frac{\dot{W}_{O_2,in,ca}}{\dot{W}_{O_2,in,ca} + \dot{W}_{N_2,in,ca} + \dot{W}_{vap,in,ca}}\\ &= x_{O_2,dry} - \frac{x_{O_2,dry,ca} - x_{O_2,out,ca}}{W_{dry,ca}/(W_{ex,ca}\cdot\alpha) + 1}\end{aligned} \tag{4}$$

$$x_{vap,in,ca} = \frac{\dot{W}_{vap,in,ca}}{\dot{W}_{O_2,in,ca} + \dot{W}_{N_2,in,ca} + \dot{W}_{vap,in,ca}} = \frac{x_{vap,out,ca}}{W_{dry,ca}/(W_{ex,ca}\cdot\alpha) + 1} \tag{5}$$

The control factors of cathode subsystem are the dry air flow rate $W_{dry,\ ca}$ and recirculation ratio α. According to this equation, concentration of oxygen will decrease as the increase of α. Dry air flow rate has no significant influence on $x_{O_2,in,ca}$, since the increase of inlet gas flow rate will cause increase of exhaust gas. And the concentration of oxygen is always lower than that without recirculation, i.e. in dry air. In Eq. (5), it is obvious that mole fraction of water increases as the increase of recirculation ratio and effect of dry air flow rate is also not significant. This is also the basic of humidification of cathodic recirculation.

For the anodic recirculation, similar theoretical Eqs. (6)–(9) can be deduced.

$$\dot{W}_{H_2,in,an} = W_{tank} + W_{re,an}\cdot x_{H_2,out,an} \tag{6}$$

$$\dot{W}_{vap,in,an} = W_{re,an}\cdot x_{vap,out,an} \tag{7}$$

$$x_{H_2,in,an} = \frac{\dot{W}_{H_2,in,an}}{\dot{W}_{H_2,in,an} + \dot{W}_{vap,in,an}} = 1 - \frac{1 - x_{H_2,out,an}}{W_{tank,an}/W_{re,an} + 1} \tag{8}$$

$$x_{vap,in,an} = \frac{\dot{W}_{vap,in,an}}{\dot{W}_{H_2,in,an} + \dot{W}_{vap,in,an}} = \frac{x_{vap,out,an}}{W_{tank,an}/W_{re,an} + 1} \tag{9}$$

Here, $W_{tank,an}$ and $W_{re,an}$ stand for flow rate of hydrogen from tank and recirculated exhaust gas from anode, respectively; $x_{i,in,an}$ and $x_{i,out,ca}$ ($i = H_2$, vap) represent molar fractions of gas species in anode inlet and outlet gas, respectively.

The control factors of anode subsystem is recirculation flux $W_{re,an}$. According to this Eqs. (8) and (9), increase of anodic recirculation flux will increase the inlet humidity of anode inlet gas, which can be reflected on the decrease of hydrogen fraction in anode inlet gas.

Voltage of fuel cell can be described as following equation, where V_{nernst} is Nernst voltage and ΔV_{ohm}, ΔV_{act} and ΔV_{trans} stand for ohmic loss, activation loss and mass transfer loss respectively.

$$V = V_{nernst} - \Delta V_{ohm} - \Delta V_{act} - \Delta V_{trans} \tag{10}$$

Nernst voltage shown in Eq. (7), where V_0 is the reversible cell potential at standard pressure. $R = 8.31\ J/(mol\cdot K)$ is the ideal gas constant and $F = 96,485\ C\cdot mol^{-1}$ is the faraday constant. T is the cell temperature in Kelvin, $p_{H_2,an}$ and $p_{O_2,ca}$ are the partial pressures of hydrogen and oxygen gases in the anode and cathode respectively while p_0 is the ambient pressure.

$$V_{nernst} = V_0 + \frac{RT}{2F}\left[ln\frac{p_{H_2,an}}{p_0} + \frac{1}{2}\ln\frac{p_{O_2,ca}}{p_0}\right] \tag{11}$$

Ohmic loss is proportional to operating current I and ohmic resistant R_{ohm}, which is usually associated with hydration of proton exchange membrane.

$$\Delta V_{ohm} = I\cdot R_{ohm} \tag{12}$$

Activation voltage loss can be calculated using Tafel equation, as shown in Eq. (9) [24], where I_n is the internal current due to fuel crossover, and I_0 is the exchange current, and α is the charge transfer coefficient.

$$\Delta V_{act} = \frac{RT}{2\alpha F}\ln\left(\frac{I + I_n}{I_0}\right) \tag{13}$$

Concentration voltage loss is caused by mass transfer, and usually appears at high current loads. It can be express as Eq. (10), according to Larminie [24], where I_1 is limiting current at which the fuel is used up at a rate equal to its maximum supply speed.

$$\Delta V_{trans} = -\frac{RT}{2F}\ln\left(1 - \frac{I}{I_1}\right) \tag{14}$$

In this study, low current operation point is studied so ohmic loss and mass transfer loss are not considered. Activation loss is cause by electrochemical reactions. Nernst voltage is associated with oxygen partial pressure and therefore can be affect through cathodic recirculation.

Therefore, in order to reduce cell voltage at low current operation point, partial pressure of oxygen needs to be reduced. Partial pressure of oxygen in cathode is not only positively associated with oxygen concentration but also with total pressure. Simply applying cathodic recirculation would lower inlet oxygen concentration but also increase total pressure in cathode. According to the analysis above, fresh air flux should be reduced together with the application of cathodic recirculation to achieve better voltage clamping effect.

Experimental study

Experimental procedure

The fuel cell stack used in this study is a 50-cell, 261 cm^2 stack with metallic bipolar plate. The specifications of the stack are shown in Table 2. As shown in Table 3, during the experiments, stack temperature is control at 68 °C. As the stack works at low current point in the experiments, air and hydrogen flow rate are set to fix value, 110 L/min and 30 L/min respectively, referenced to the recommended minimum value of the stack.

Two experiments are carried out. In the first experiment, the fuel cell stack works at a constant current of 31 A (119 mA/cm^2), the inlet air is humidified and the inlet hydrogen is not humidified. Anode is set to dead-end mode, and purge valve opens for 600 ms each 15 s. Then, the membrane humidifier on the cathode side is turned off at t0, while the current is unchanged. After the voltage became stable, cathodic recirculation is applied by turning on the cathodic recirculation pump. At the same time, effect of anodic recirculation is also study during the test. Variations of current, average cell voltage, pump speed and humidity of inlet air are presented in Fig. 3.

In the second experiment, inlet fresh air flow rate and cathodic recirculation are applied simultaneously to study the voltage clamping effect of cathodic recirculation. In this experiment, fuel cell stack works at constant value of 31 A (119 mA/cm^2), the speed of cathodic recirculation pump increases from 0 to 3000 r min^{-1}, and the flow rate of fresh air decreases from 110 to 20 L/min. Variation of current, average cell voltage, pump speed and concentration of oxygen in inlet air are presented in Fig. 5. In order to investigate the voltage reduction effect, same experiment is performed at another operation point, 14 A (54 mA/cm^2).

Results and discussion

As shown in Fig. 3(b), voltage drops down gradually during the time period of [t_0, t_1], because inlet air is not humidified and the relative humidity of air in the stack decreases gradually. At the timing t_1, cathodic recirculation pump is turned on, and the inlet gas is the mixture of fresh dry air and exhaust air. Relative humidity rises up rapidly from 42% to 60% in 5 s, and then increases gradually to 98% in 100 s. The humidified effect is similar to that with an external humidifier. During this period, the average cell voltage decreases slightly, and then increases from 0.776 V to a stable value 0.779 V. It is 0.016 V smaller than that with an external humidifier, 0.795 V. The anode purging valves opens every 15 s, causing slightly voltage drops, as shown in Fig. 3(b). During the cathodic recirculation operation, the anodic recirculation pump is turned on at the timing t_2, and turned off at the timing t_3. It can be seen that, due to the opening of the anodic pump, the cell voltage drops slightly, and the relative humidity of inlet air drops from 98% to about 90%.

Table 2 – Specifications of fuel cell stack.

Parameters	Values
Cells number	50
Area	261 cm^2
Rated operation point	261 A, 30 V
Max power	10 kW
Operating temperature	60–70 °C

Table 3 – Experiments condition.

Parameters	Values
Pressure at anode outlet (atm)	1
Pressure at cathode outlet (atm)	1
Flow rate of inlet H_2 (L/min)	30
Flow rate of inlet fresh air (L/min)	20–110
Coolant flow rate (L/min)	22
Temp. of coolant inlet (°C)	68

Fig. 4(a, b) illustrates the oxygen and hydrogen concentrations of inlets and outlets on the cathode and anode side. Neglecting nitrogen in anode inlet and outlet gas, vapor concentration can be calculated based on hydrogen concentration, as shown in (b) in Fig. 4. When anodic recirculation turns on at t_2, 1) hydrogen molar fraction in the anode inlet decreases, which means that the molar fraction of water in the anode inlet increases; 2) hydrogen molar fraction in the anode outlet increases, which means that the water molar fraction in the anode outlet decreases; 3) oxygen water molar fraction both in the cathode inlet and outlet increases, which means that the water molar fraction in the cathode side decreases.

It indicated that, when the anode recirculation pump is turned on, the water molar fraction on the cathode side and on the anode outlet decreases. This can be explained as follows. 1) The reduction of relative humidity on the cathode side is mainly due to the water transport through the membrane. When the anode recirculation pump is turned on, some water flux goes through the membrane from cathode to anode, leading to a slightly reduction in the water content on the cathode side. 2) When the anode recirculation pump is off before t_2, water accumulates at the anode outlet. When it is turned on, the water goes out of the anode outlet, leading a reduction in the water content in anode.

The first experiment shows that, cathodic recirculation is effective in humidifying the inlet air, but the cell voltage can't be reduced significantly when fresh air flow rate remains unchanged. It decreases from 0.795 V to 0.779 V when the current density is 119 mA/cm^2 and the recirculation pump turns on. When cathodic and anodic recirculation are applied together, humidity of cathode inlet gas is a little lower than the case of only cathodic recirculation, and both cathode and anode inlet gas are humidified.

As mention in Section Theoretical analysis, fresh air flow rate should be reduced during cathodic recirculation to achieve a larger voltage drop. In the second experiment, the anodic recirculation pump keeps off and the cathodic recirculation pump is turned on. Both the rotational speed of the cathodic pump and the fresh air flow changes, the dynamic process is illustrated in Fig. 5. The cell current remains unchanged at 31 A (119 mA/cm^2). The fresh air flow decreases from 110 L/min to 40 L/min gradually. At the same time, the cathodic recirculation pump is turn on and the speed is

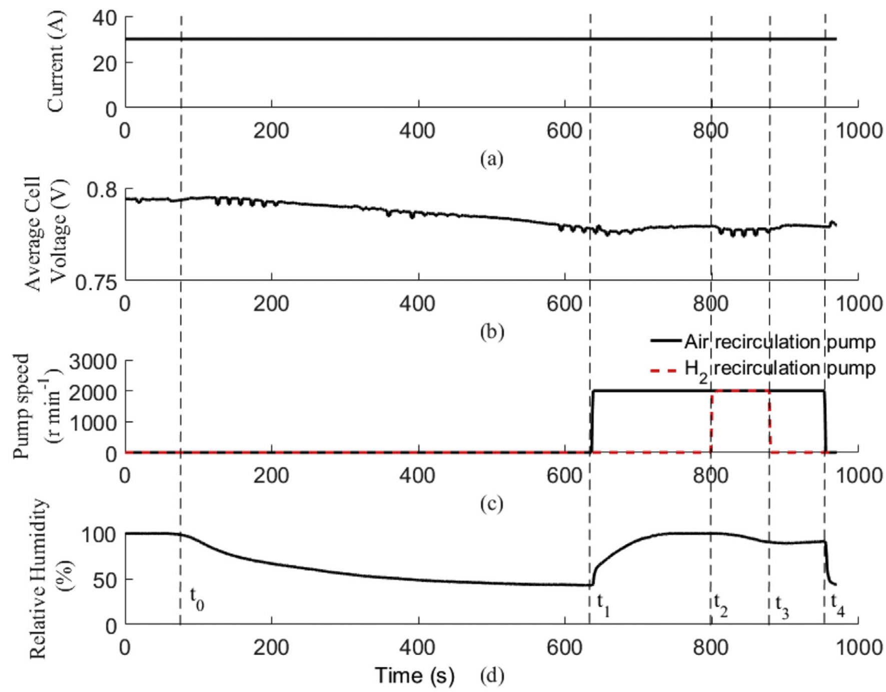

Fig. 3 – Self-humidification through cathodic recirculation.

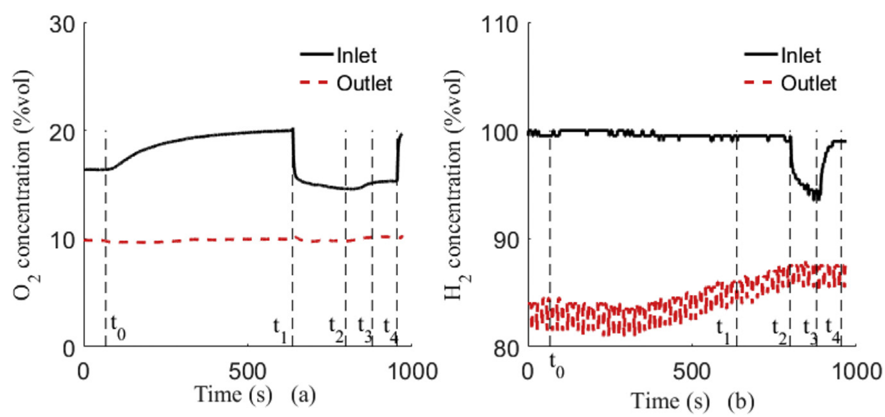

Fig. 4 – Variation of oxygen and hydrogen concentration during experiment 1.

adjusted from 0 to 3000 r min^{-1} gradually. As a result, the oxygen concentration at cathode inlet drops from 18.2% to 9.8%, the oxygen concentration at outlet also drops from 13.7% to 4.9%, and the average cell voltage decreases from 0.772 V to 0.732 V. About a voltage drop of 0.040 V is observed. It is significantly larger than the voltage drop of 0.016 V in the first experiment.

Same operation is performed at 14 A (54 mA/cm^2), as shown in Fig. 6. The fresh air flow decreases from 110 L/min to 20 L/min gradually. At the same time, the cathodic recirculation pump is turn on and the speed is adjusted from 0 to 3000 r min^{-1} gradually. As a result, the oxygen concentration at cathode inlet drops from 18.6% to 7.2% while the oxygen concentration at outlet also drops from 15.8% to 4.1%. Average cell voltage drop of 0.042 V is observed, from 0.828 V to 0.780 V.

In this study, the nominal SR_n (stoichiometric ratio) is defined in Eq. (15) to describe the amount of fresh air into the fuel cell stack, where $W_{dry,ca}$ is fresh air flow rate in L/min under standard state (273.15 K, 101.325 kPa), I is the current and $F = 96{,}485$ C/mol is the faraday constant. $n = 4$ stands for the number of electron in one oxygen molecule. $N_{cell} = 50$ is the cells number of the stack in this study.

$$SR_n = \frac{W_{dry,ca} \cdot 0.21}{I/F/n \cdot N_{cell} \cdot 22.4 \cdot 60} \tag{15}$$

At 31 A, SR_n decreases from 4.25 to 1.52 when fresh air flux

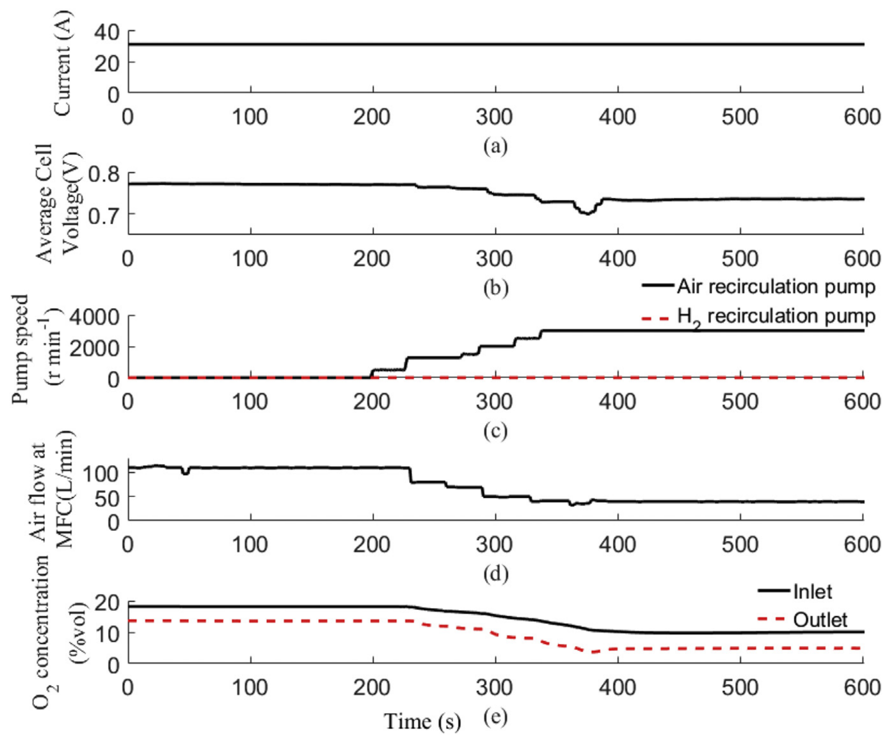

Fig. 5 – Voltage clamping through cathodic recirculation at *I* = 31 A.

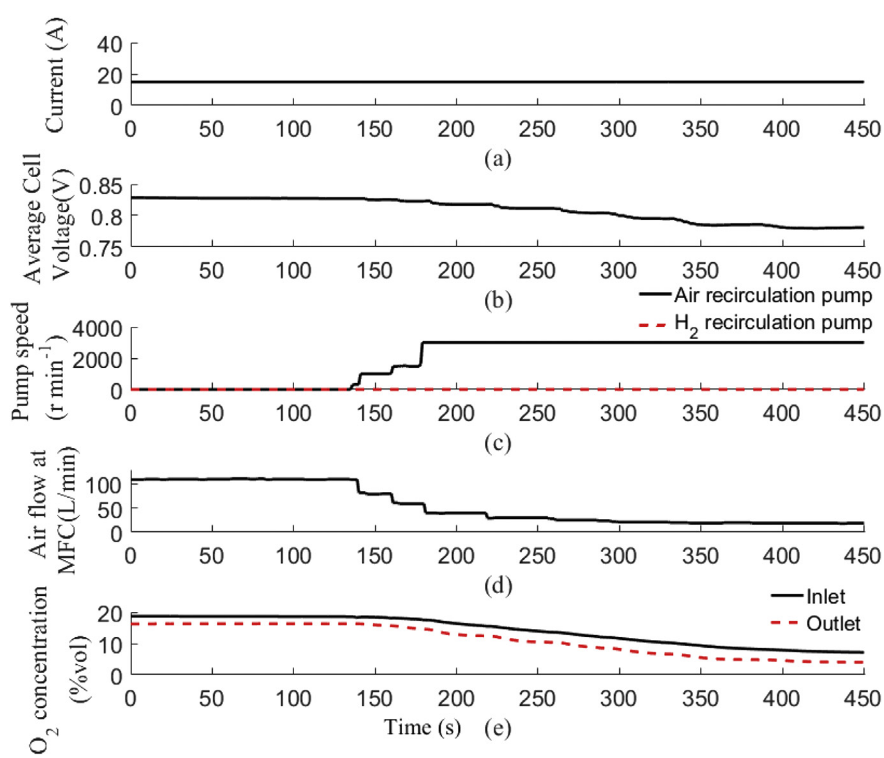

Fig. 6 – Voltage clamping through cathodic recirculation at *I* = 14 A.

decreases from 110 to 40 L/min, while it decreases from 9.5 to 1.64 at 14 A when air flux decreases from 110 to 20 L/min. Variations of average cell voltage at different SR value are shown in Fig. 7. Although voltages are different at two different operation conditions, the curves are similar. Fig. 7 shows a strong correlation between SR_n and average cell voltage. Especially when SR is less than 2.5, the voltage reduction becomes more obvious with the decrease of SR_n. At

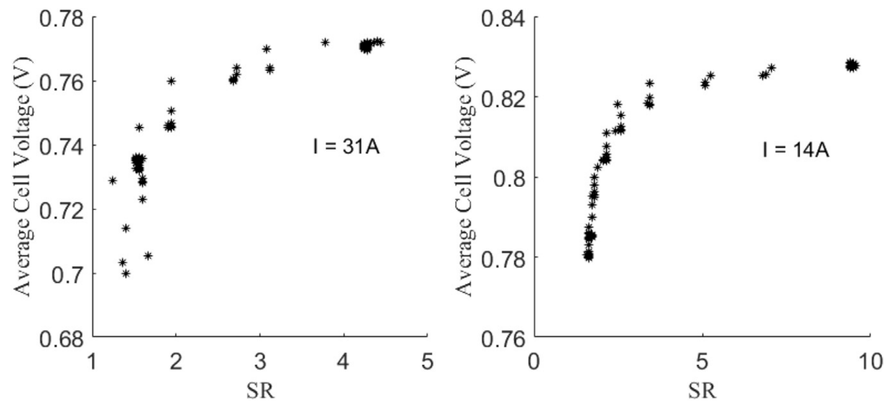

Fig. 7 – Relationship between the nominal stoichiometry ratio (SR_n) and average cell voltage at $I = 31$ A and $I = 14$ A.

31 A, the voltage drop is 0.012 V (from 0.776 to 0.760 V) when SR varies from 4.25 to 2.68, while it drops about 0.028 V (from 0.760 to 0.732 V) when SR varies from 2.68 to 1.52. At 14 A, voltage drop from SR = 2.6 to 1.64 is 0.03 V, much larger than that from SR = 9.5 to 2.6, 0.18 V.

Conclusion

This paper presents an experimental study on a PEM fuel cell system with anodic and cathodic recirculation. A fuel cell system with cathodic and anodic recirculation is built up based on a 10 kW fuel cell stack, which consists of 50 cells and has an active area of 261 cm^2. Key parameters, e.g., stack current, stack voltage, cell voltage, air flow, relative humidity on the cathode side, oxygen concentration at the inlet and outlet of the cathode side, are measured. Theoretical analysis shows that, inlet air humidity can be improved by cathodic recirculation and the water concentration increases with the increase of recirculation ratio. Fresh air flow rate reduction should be applied together with the cathode recirculation to get better voltage clamping effect.

Experimental study based on the built up system is carried out. Relative humidity of inlet gas can reach 98% by using cathodic recirculation. It indicates that, with a cathodic recirculation the system gets good self-humidification effect, which is similar to that with an external humidifier. When both the cathodic and anodic recirculation operate, the relative humidity of hydrogen on the anode side increases, and the relative humidity of mixed air on the cathode side decreases, compared to the situation with only cathodic recirculation. The different trends on the anode and cathode side are mainly due to the water transport through the membrane. When the anode recirculation pump is turned on, some water flux goes through the membrane from cathode to anode, which leads to a slightly reduction in the relative humidity on the cathode side. By applying cathodic recirculation and reducing the fresh air flux meanwhile, the average cell voltage drop becomes more obviously. In this study, voltage drop due to cathodic EGR increases from 0.016 V (@119 mA/cm^2 and constant fresh air flux) to 0.040 V (@119 mA/cm^2 and reduced fresh air flux) and 0.042 V (@54 mA/cm^2 and reduced fresh air flux). Coloration between the nominal stoichiometric ratio (SR_n) and average cell voltage indicates that, using this voltage clamping method, voltage reduction will be notable when the SR_n is less than 2.5.

Dynamic model of fuel cell system with dual recirculation can be built and used to investigate self-humidification and voltage clamping based on the experimental study in this paper. Advanced control method of dual recirculation system is also needed. Further study will focus on the dynamic model and control of the dual recirculation fuel cell system.

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REFERENCES

[1] Alaswad A, Baroutaji A, Achour H, Carton J, Al Makky A, Olabi AG. Developments in fuel cell technologies in the transport sector. Int J Hydrogen Energy 2015;41:16499–508.

[2] Veziroglu A, Macario R. Fuel cell vehicles: state of the art with economic and environmental concerns. Int J Hydrogen Energy 2011;36:25–43.

[3] Li J, Hu Z, Xu L, Ouyang M. Fuel cell system degradation analysis of a Chinese plug-in hybrid fuel cell city bus. Int J Hydrogen Energy 2016;41:15295–310.

[4] Wang Y, Chen KS, Mishler J, Cho SC, Adroher XC. A review of polymer electrolyte membrane fuel cells: technology, applications, and needs on fundamental research. Appl Energy 2011;88:981–1007.

[5] Borup R, Meyers J, Pivovar B, Kim YS, Mukundan R, Garland N, et al. Scientific aspects of polymer electrolyte fuel cell durability and degradation. Chem Rev 2007;107:3904–51.

[6] Schmittinger W, Vahidi A. A review of the main parameters influencing long-term performance and durability of PEM fuel cells. J Power Sources 2008;180:1–14.
[7] Wu J, Yuan XZ, Martin JJ, Wang H, Yang D, Qiao J, et al. Proton exchange membrane fuel cell degradation under close to open-circuit conditions. Part I: in situ diagnosis. J Power Sources 2010;195:1171–6.
[8] Yoon W, Huang X. Study of polymer electrolyte membrane degradation under OCV hold using bilayer MEAs. J Electrochem Soc 2010;157:B599–606.
[9] Noto H, Kondo M, Otake Y, Kato M. Development of fuel cell hybrid vehicle by Toyota-durability. Hybrid technology. SAE Tech Pap 2009:1–6.
[10] Kim H-Y, Kim K. Numerical study on the effects of gas humidity on proton-exchange membrane fuel cell performance. Int J Hydrogen Energy 2016:4–11.
[11] He J, Choe S-Y, Hong C-O. Modelling and control strategy of a fuel delivery system with dual recirculation lines for a PEM fuel cell system. In: 6th int. fuel cell sci. eng. technol. conf; 2008. p. 661–73.
[12] Migliardini F, Di Palma TM, Gaele MF, Corbo P. Hydrogen purge and reactant feeding strategies in self-humidified PEM fuel cell systems. Int J Hydrogen Energy 2016;42:1–8.
[13] Karnik AY, Sun JSJ, Buckland JH. Control analysis of an ejector based fuel cell anode recirculation system. In: 2006 Am control conf; 2006. p. 484–9.
[14] Bao C, Ouyang M, Yi B. Modeling and control of air stream and hydrogen flow with recirculation in a PEM fuel cell system-II. Linear and adaptive nonlinear control. Int J Hydrogen Energy 2006;31:1897–913.
[15] Bao C, Ouyang M, Yi B. Modeling and control of air stream and hydrogen flow with recirculation in a PEM fuel cell system-I. Control-oriented modeling. Int J Hydrogen Energy 2006;31:1879–96.
[16] Uno M, Shimada T, Tanaka K. Reactant recirculation system utilizing pressure swing for proton exchange membrane fuel cell. J Power Sources 2011;196:2558–66.
[17] Uno M, Shimada T, Tanaka K. Reactant recirculation technique utilizing pressure swing for proton exchange membrane fuel cell system. In: 8th annu int energy convers eng conf; 2010.
[18] Nonobe Y, Hasegawa T, Imanishi H, Nada M, Ikogi Y. Development of the fuel cell system in the Mirai FCV. SAE Tech Pap 2016;12:5.
[19] Breault RD. High temperature reactant recycling for PEM fuel cell humidification. US 7067209 B2. 2006.
[20] Hild T, Herbig T, Wnendt B. Cathode humidification of a PEM fuel cell through exhaust gas recirculation into a positive displacement compressor. US 7781084 B2. 2010.
[21] Kim BJ, Kim MS. Studies on the cathode humidification by exhaust gas recirculation for PEM fuel cell. Int J Hydrogen Energy 2012;37:4290–9.
[22] Hu J, Xu L, Li J, Ouyang M, Cheng S, Fang C. Water management in a self-humidifying PEM fuel cell system by exhaust gas recirculation. In: 2014 IEEE conf expo transp electrif Asia-Pacific (ITEC Asia-Pacific), 1; 2014. p. 1–6.
[23] Cheng S, Li J, Xu L, Ouyang M. Air supply system model with exhaust gas recirculation for improving the life of fuel cell. In: IEEE transp electrif conf expo, ITEC Asia-Pacific 2014-conf proc; 2014. p. 1–6.
[24] Larminie J, Dicks A, McDonald MS. Fuel cell systems explained, vol. 2; 2003.